

HORNSBY GIRLS HIGH SCHOOL



Mathematics Extension 1

Year 12 Higher School Certificate
Trial Examination Term 3 2018

STUDENT NUMBER: _____

General Instructions

- Reading Time – 5 minutes.
- Working Time – 2 hours.
- Write using black or blue pen.
Black pen is preferred.
- NESA-approved calculators and drawing templates may be used.
- A reference sheet is provided separately.
- In Questions 11 – 14, show relevant mathematical reasoning and/or calculations.
- Marks may be deducted for untidy and poorly arranged work.
- Do not use correction fluid or tape.
- Do not remove this paper from the examination room.

Total marks – 70

Section I Pages 3 – 6

10 marks

Attempt Questions 1 – 10

Answer on the Objective Response Answer Sheet provided.

Section II Pages 7 – 14

60 marks

Attempt Questions 11 – 14.

Start each question in a new writing booklet.

Write your student number on every writing booklet.

Question	1-10	11	12	13	14	Total
Total	/10	/15	/15	/15	/15	/70

This assessment task constitutes 45% of the Higher School Certificate Course School Assessment

Section I

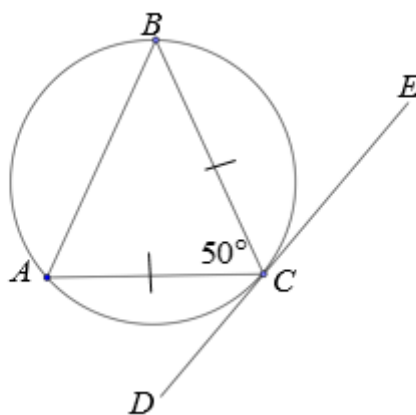
10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the Objective Response answer sheet for Questions 1 – 10

- 1 A, B and C lie on a circle with $AC = BC$, DE is a tangent to the circle at C .
The size of $\angle ACB = 50^\circ$. What is the size of $\angle ACD$?



NOT TO
SCALE

- (A) 50°
(B) 60°
(C) 65°
(D) 80°
- 2 Which expression is equal to $\sqrt{3} \sin x - \cos x$?

- (A) $2 \sin \left(x - \frac{\pi}{6} \right)$
(B) $2 \sin \left(x + \frac{\pi}{6} \right)$
(C) $\sqrt{10} \sin \left(x - \frac{\pi}{6} \right)$
(D) $\sqrt{10} \sin \left(x + \frac{\pi}{6} \right)$

3 What is the domain of the function $y = 3\sin^{-1}(2 - x)$?

- (A) $0 \leq x \leq 2$
- (B) $3 \leq x \leq 9$
- (C) $1 \leq x \leq 3$
- (D) $-3 \leq x \leq -1$

4 What is the value of the obtuse angle between the lines $5x + 2y - 3 = 0$ and $y = 2x - 1$?

- (A) $48^\circ 22'$
- (B) $175^\circ 14'$
- (C) $150^\circ 21'$
- (D) $131^\circ 38'$

5 Which group of three numbers could be the roots of the polynomial equation below?

$$x^3 + kx^2 - 17x - 60 = 0$$

- (A) $-2, 5, 6$
- (B) $1, 2, -30$
- (C) $-3, 4, -5$
- (D) $-1, 1, 60$

6 What is the derivative of $4 \tan^{-1} \frac{x}{2}$?

(A) $\frac{4}{4+x^2}$

(B) $\frac{8}{4+x^2}$

(C) $\frac{16}{1+x^2}$

(D) $\frac{16}{4+x^2}$

7 The inverse function of $y = \frac{4}{x^2}$, $x > 0$, is:

(A) $y = \frac{2}{\sqrt{x}}$

(B) $y = \frac{2}{x}$

(C) $y = -\frac{2}{x}$

(D) $y = \pm \frac{2}{\sqrt{x}}$

8 A committee of six is to be formed from 7 boys and 5 girls. The committee has to contain at least 3 girls. In how many ways can the committee be chosen?

(A) 210

(B) 350

(C) 462

(D) 924

- 9 When $P(x) = x^4 + 6x^3 - 5x^2 + 7$ is divided by $x^2 - 1$, the remainder is $6x + k$.

What is the value of k ?

- (A) -9
- (B) -3
- (C) 3
- (D) 6

- 10 Given $4\sin^2 18^\circ + 1 = \sqrt{A}$, what is the value of $\cos 36^\circ$ in terms of A ?

- (A) $\frac{3 - \sqrt{A}}{2}$
- (B) $\frac{1 - A}{2}$
- (C) $\frac{\sqrt{A} + 1}{4}$
- (D) $\frac{\sqrt{A}}{4} - 1$

End of Section I

Section II

60 marks

Attempt Questions 11 – 14

Allow about 1 hour and 45 minutes for this section

Answer each question in a new writing booklet. Extra writing booklets are available.

In Questions 11 – 14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start a new writing booklet

(a) Solve $\frac{x^2}{x+12} \leq 1$. 3

(b) Sketch the graph $y = 2 \tan^{-1} x + \pi$, clearly indicating all asymptotes. 2

(c) Evaluate $\int_{0.5}^{1.5} \frac{1}{\sqrt{2x-x^2}} dx$, using the substitution $x = u + 1$, giving your answer 3
in simplest exact form.

(d) Differentiate $\frac{x \log_e x}{e^x}$. 2

(e) Find $\lim_{x \rightarrow 1} \frac{1 - \frac{1}{x^3}}{1 - \frac{1}{x^2}}$. 3

(f) Solve for x : $\frac{1}{x-1} - 2 = \frac{-3}{x+1}$. 2

End of Question 11

Question 12 (15 marks) Start a new writing booklet

(a) Evaluate $\lim_{x \rightarrow 0} \frac{\sin 4x}{7x}$. **1**

(b) Find the volume of the solid of revolution formed when the area bounded by the curve $y = \sin 2x$ and the x -axis is rotated about the x -axis from $x = 0$ to $x = \frac{\pi}{4}$. **3**

(c) The velocity of a particle is $v = 5x + 3 \text{ ms}^{-1}$. **3**
If the initial displacement is 2 m, to the right of the origin, show that the displacement as a function of time is $x = \frac{1}{5}(13e^{5t} - 3)$.

(d) A dinner party is arranged for 16 people. The people will be seated along two sides of a rectangular table with 8 chairs on each side. Four people wish to sit on one particular side and a couple wishes to sit on the other side next to each other. **2**

In how many ways can the 16 people be seated?

Question 12 continues on page 9

Question 12 (continued)

- (e) A particle is moving in simple harmonic motion about the origin, with displacement x metres. The displacement is given by $x = 2 \cos\left(3t + \frac{\pi}{3}\right)$.

(i) What is the total distance travelled by the particle when it first reaches the origin? 2

(ii) Find the maximum speed of the particle. 1

- (f) A tub of ice cream taken out of the freezer has a temperature of -18°C . It is placed in a room of constant temperature 23°C . 3

After t minutes the temperature, $T^{\circ}\text{C}$ of the ice cream is given by:

$$T = 23 - Ae^{-0.04t},$$

where A is a positive constant.

How long does it take for the ice cream to reach a temperature of -12°C which is considered to be the ideal temperature for serving ice cream?

End of Question 12

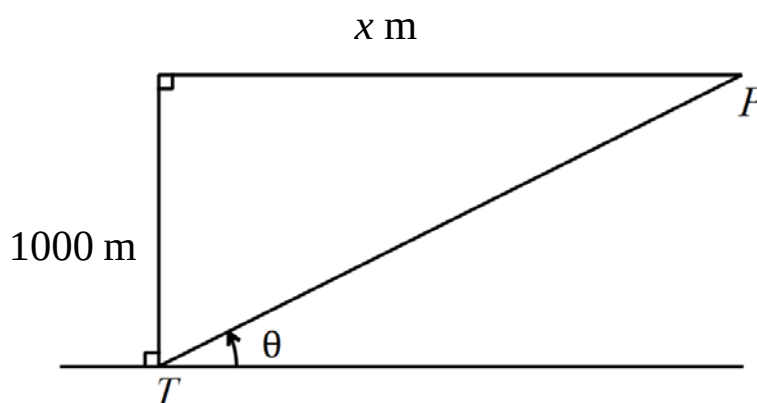
Question 13 (15 marks) Start a new writing booklet.

- (a) Use mathematical induction to prove that $n^3 + 2n$ is divisible by 3 for all integers $n \geq 1$.

3

- (b) Tom, on horizontal ground, is looking at an aeroplane P through a telescope T .

The aeroplane is approaching at a speed of 200 ms^{-1} at a constant altitude of 1000 m above the telescope. When the horizontal distance of the aeroplane from the telescope is $x \text{ m}$, the angle of elevation of the aeroplane is θ radians.



NOT TO
SCALE

- (i) Show that $\frac{d\theta}{dt} = \frac{2 \times 10^5}{x^2 + 10^6}$.

2

- (ii) Find the rate at which θ is changing when $\theta = \frac{\pi}{4}$, giving your answer to the nearest degree per second.

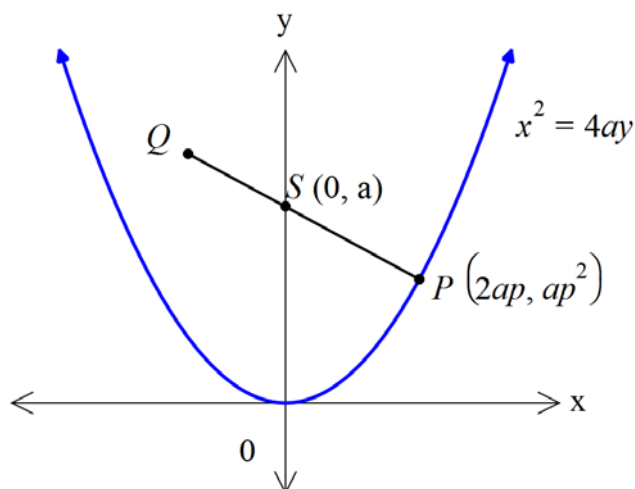
2

Question 13 continues on page 11

Question 13 (continued)

- (c) In the diagram below, $P(2ap, ap^2)$ is a point on the parabola $x^2 = 4ay$ with focus $S(0, a)$.

The point Q lies on PS produced such that Q divides PS externally in the ratio 3:1.



NOT TO
SCALE

- (i) Show that Q has coordinates $\left(-ap, \frac{3a - ap^2}{2}\right)$. 1

- (ii) Show that as P varies, the locus of Q is a parabola with equation 2

$$x^2 = -2a\left(y - \frac{3a}{2}\right).$$

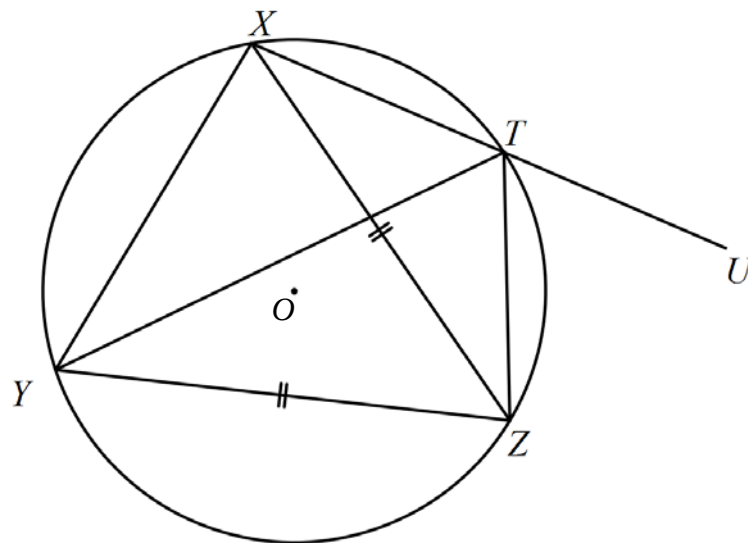
- (iii) Give a geometrical description of the locus of Q , stating all important features. 2

Question 13 continues on page 12

Question 13 (continued)

- (d) In the diagram below, XYZ is a triangle in which $XZ = YZ$. T is a point on the minor arc XZ of the circle centre O , passing through X , Y and Z . XT is produced to U .

3



NOT TO
SCALE

Copy or trace the diagram into your writing booklet.

Prove that TZ bisects $\angle UTY$.

End of Question 13

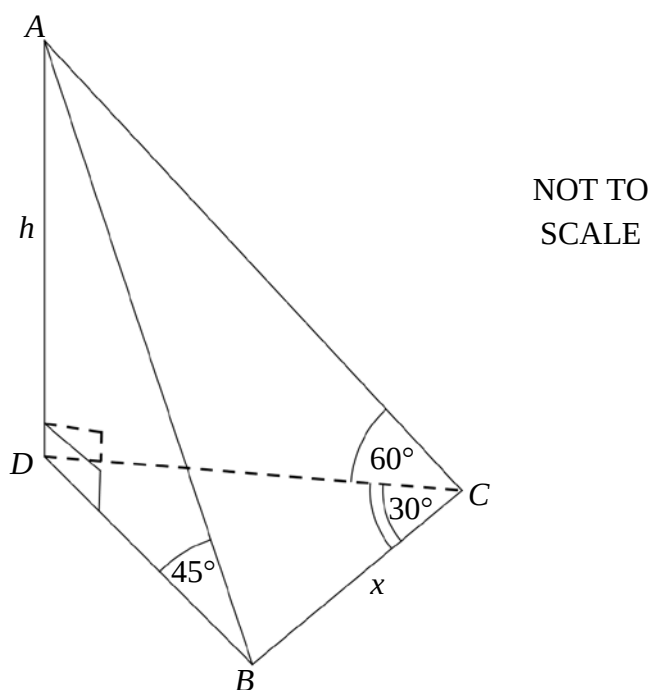
Question 14 (15 marks) Start a new writing booklet

- (a) (i) Write down the general solutions to the equation $2 \cos 3\theta = 1$. 2

- (ii) Hence, or otherwise, find all solutions of the equation 2

$$2 \cos 3\theta = 1 \text{ for } 0 \leq \theta \leq \pi.$$

- (b) In the diagram below, $ABCD$ is a triangular pyramid with base $\triangle BCD$ and perpendicular height $AD = h$. $\angle BCD = 30^\circ$, $\angle ABD = 45^\circ$ and $\angle ACD = 60^\circ$.



- (i) Use the cosine rule to show that $2h^2 + 3xh - 3x^2 = 0$. 3

- (ii) Hence show that $\frac{h}{x} = \frac{\sqrt{33}-3}{4}$. 2

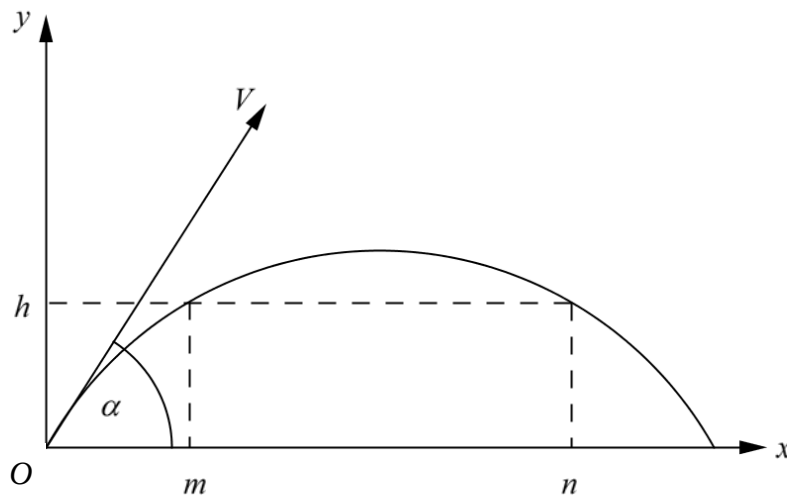
Question 14 continues on page 14

Question 14 (continued)

- (c) An object is projected with velocity $V \text{ ms}^{-1}$ from a point O at an angle of elevation α . Axes x and y are taken horizontally and vertically through O respectively.

The object just clears two vertical chimneys of height h meters at horizontal distance m metres and n metres from O .

The acceleration due to gravity is taken as 10 ms^{-2} and air resistance is ignored.



NOT TO
SCALE

- (i) Show that the expressions of the particle's horizontal and vertical displacements after t seconds are given by

2

$$x = Vt \cos \alpha \text{ and } y = Vt \sin \alpha - 5t^2.$$

- (ii) Show that at $x = m$, $V^2 = \frac{5m^2(1 + \tan^2 \alpha)}{m \tan \alpha - h}$.

2

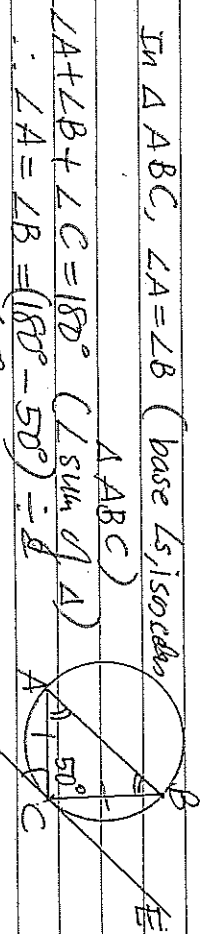
- (iii) Show that $\tan \alpha = \frac{h(m+n)}{mn}$.

2

End of Paper

Section I

1) In $\triangle ABC$, $\angle A = \angle B$ (base $\angle$ s, isosceles)



$$\angle A + \angle B + \angle C = 180^\circ \quad (\text{sum of } \triangle)$$

$$\therefore \angle A = \angle B = (180^\circ - 50^\circ) \div 2 = 65^\circ$$

$$\angle ACD = \angle B \quad (\angle \text{ in alternate segment})$$

$$= 65^\circ$$

Ans. C

2) $\sqrt{3} \sin x - \cos x = 2 \quad \sqrt{(\sqrt{3})^2 + 1^2} = 2$

$$= 2 \left(\frac{\sqrt{3}}{2} \sin x - \frac{1}{2} \cos x \right)$$

$$= 2 \sin(x - \alpha) \quad \text{where } \tan \alpha = \frac{1/2}{\sqrt{3}/2}$$

$$= 2 \sin(x - \frac{\pi}{6}) \quad = \frac{1}{\sqrt{3}}$$

Ans. A

$$\alpha = \frac{\pi}{6}$$

3) $y = 3 \sin^{-1}(2-x)$

$$D: -1 \leq (2-x) \leq 1$$

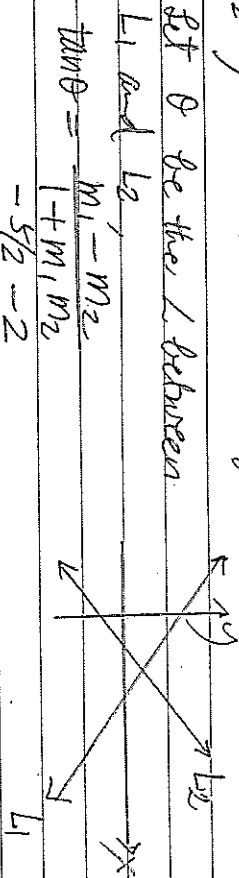
$$-3 \leq -x \leq -1$$

$$\therefore 3 \geq x \geq 1$$

$$1 \leq x \leq 3$$

Ans. C

4) $L_1: 5x + 2y - 3 = 0$, gradient $= -\frac{5}{2} = m_1$
 $L_2: y = 2x - 1$, gradient $= 2 = m_2$



Let θ be the $\angle$ between L_1 and L_2 ,

$$\tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2}$$

$$= \frac{-\frac{5}{2} - 2}{1 + (-\frac{5}{2})(2)}$$

$$= \frac{-\frac{9}{2}}{-8} = \frac{9}{16}$$

$\theta = 48^\circ 22' \text{ (nearest min.)}$
 the obtuse $\angle = 180^\circ - 48^\circ 22' = 131^\circ 38'$

Ans. D

5) Let α, β, γ be the roots of $x^3 + px^2 - 17x + 60 = 0$

$$\alpha\beta\gamma = -(-60)/1 = 60$$

$$\text{In } (C), (-3)(4)(-5) = 60$$

Ans. C

6) $\frac{d}{dx} 4 \tan^{-1} \frac{x}{2} = 4 \times \frac{1}{1 + (\frac{x}{2})^2} \times \frac{1}{2}$

$$= 4 \times \frac{1}{4 + x^2} \times \frac{1}{2}$$

$$= \frac{2}{4 + x^2}$$

Ans. B

P.3

$$7) \quad y = \frac{4}{x^2}, \quad x > 0$$

To find the inverse, interchange x and y ,

$$x = \frac{4}{y^2}, \quad y > 0$$

$$y^2 = \frac{4}{x}$$

$$y = \pm \sqrt{\frac{4}{x}} = \pm \frac{2}{\sqrt{x}}$$

Since $y > 0$, $y = \frac{2}{\sqrt{x}}$ is the inverse.

Ans. A

8) From 7 boys and 5 girls, a committee of 6 is to be formed. The committee has to have at least 3 girls.

Number of ways

$$\begin{aligned} &= \text{ways to have 3 girls and 3 boys} \\ &+ \text{ways to have 4 girls and 2 boys} \\ &+ \text{ways to have 5 girls and 1 boy} \\ &= {}^5C_3 \times {}^7C_3 + {}^5C_4 \times {}^7C_2 + \\ &= {}^{350} + {}^{105} + 7 \\ &= 462 \end{aligned}$$

Ans. C.

9) When $P(x) = x^4 + 6x^3 - 5x^2 + 7$ is divided by $x^2 - 1$, the remainder is $6x + k$.

$P(x) = (x^2 - 1)Q(x) + 6x + k$, where $Q(x)$ is the quotient.

$$\begin{aligned} P(1) &= (1^2 - 1)Q(1) + 6 + k \\ 1 + 6 + 1 - 5 + 7 &= 6 + k \end{aligned}$$

$$k = 3$$

Ans. C

P.4

$$10) \quad 4 \sin^2 18^\circ + 1 = \sqrt{A}$$

$$2 \sin^2 18^\circ = \frac{1}{2}(\sqrt{A} - 1)$$

$$\cos 36^\circ = 1 - 2 \sin^2 18^\circ \quad (\text{By } \cos 2\theta = 1 - 2 \sin^2 \theta, \text{ and } \theta = 18^\circ)$$

$$= 1 - \frac{1}{2}(\sqrt{A} - 1)$$

$$= 1 - \frac{1}{2}\sqrt{A} + \frac{1}{2}$$

$$= \frac{3}{2} - \frac{1}{2}\sqrt{A}$$

Ans. A.

Section II

P.5

Question 1

(a) $\frac{x^2}{x+12} \leq 1, \quad x \neq -12$

$$x^2(x+12) \leq (x+12)^2$$

$$x^2(x+12) - (x+12)^2 \leq 0$$

$$(x+12)\{x^2 - (x+12)\} \leq 0$$

$$(x+12)(x^2 - x - 12) \leq 0$$

$$(x+12)(x-4)(x+3) \leq 0$$

$$x \leq -12 \quad \text{or} \quad -3 \leq x \leq 4$$

Since $x \neq -12$,

$$x < -12 \quad \text{or} \quad -3 \leq x \leq 4$$

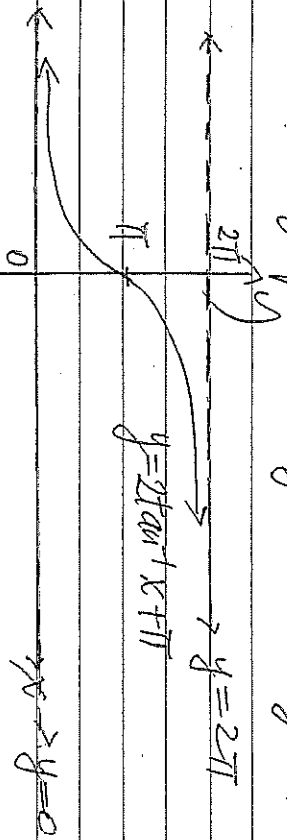
(b) $-\frac{\pi}{2} < \tan^{-1}x < \frac{\pi}{2}$

$$-\pi < 2\tan^{-1}x < \pi$$

$$0 < 2\tan^{-1}x + \pi < 2\pi$$

for $y = 2\tan^{-1}x + \pi$,
 $0 < y < 2\pi$

When $x = 0$, $\tan^{-1}x = 0$, $y = \pi$
 Horizontal asymptotes: $y = 0$ and $y = 2\pi$.



(c) $\int_{0.5}^{1.5} \frac{1}{\sqrt{2x-x^2}} dx$, Put $x = u + 1$
 $dx = du$

$= \int_{0.5}^{1.5} \frac{1}{\sqrt{1-u^2}} du$
 When $x = 0.5$, $u = -0.5$
 When $x = 1.5$, $u = 0.5$
 $2x - x^2 = x(2-x)$

$= [\sin^{-1}u]_{-0.5}^{0.5}$
 $= (u+1)\sqrt{1-u^2}$
 $= (u+1)(1-u)$
 $= 1-u^2$

$= \sin^{-1}0.5 - \sin^{-1}(-0.5)$

$= \frac{\pi}{6} - (-\frac{\pi}{6})$

$= \frac{\pi}{3}$

(d) $\frac{d}{dx} x \ln x = e^x \frac{d}{dx} (x \ln x) - (x \ln x) \frac{d}{dx} e^x$

$= e^x (x \times \frac{1}{x} + 1 \times \ln x) - (x \ln x) x^x$
 $(e^x)^2$

$= \frac{1 + \ln x - x \ln x}{e^x}$

(e) $\lim_{x \rightarrow 1} \frac{1 - x^3}{1 - x^2} = \lim_{x \rightarrow 1} \frac{x^3 - 1}{x(x^2 - 1)}$ (multiply by $\frac{x^3}{x^3}$)

$= \lim_{x \rightarrow 1} \frac{(x-1)(x^2+x+1)}{x(x-1)(x+1)}$

$= \lim_{x \rightarrow 1} \frac{x^2+x+1}{x(x+1)}$

$= \frac{1^2+1+1}{1(1+1)}$

$= \frac{3}{2}$

P.6

P.7

$$(f) \frac{1}{x-1} - 2 = \frac{-3}{x+1}, \quad x \neq -1, -1.$$

Multiply by $(x-1)(x+1)$,

$$(x+1) - 2(x-1)(x+1) = -3(x-1)$$

$$x+1 - 2(x^2-1) + 3x-3=0$$

$$x+1 - 2x^2 + 2 + 3x - 3 = 0$$

$$2x^2 - 4x = 0$$

$$2x(x-2) = 0$$

$$\therefore x=0 \text{ or } 2$$

Question 12

P.8

$$(a) \lim_{x \rightarrow 0} \frac{\sin 4x}{7x} = \lim_{x \rightarrow 0} \frac{4}{7} \frac{\sin 4x}{4x}$$

$$= \frac{4}{7} \lim_{x \rightarrow 0} \frac{\sin 4x}{4x}$$

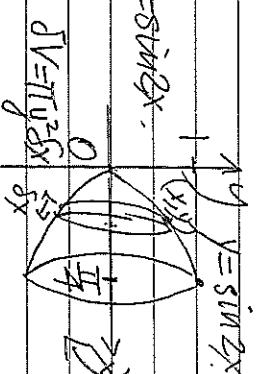
$$= \frac{4}{7} \times 1$$

$$= \frac{4}{7}$$

(b) Volume of the solid

$$= \int_0^{\pi/4} \pi y^2 dx \text{ where } y = \sin 2x.$$

$$= \pi \int_0^{\pi/4} \sin^2 2x \cdot dx$$



$$= \pi \int_0^{\pi/4} \frac{1 - \cos 4x}{2} dx \quad (\text{By } \sin^2 \theta = \frac{1 - \cos 2\theta}{2}, \theta = 2x)$$

$$= \frac{\pi}{2} \left[x - \frac{1}{4} \sin 4x \right]_0^{\pi/4}$$

$$= \frac{\pi}{2} \left[\frac{\pi}{4} - \frac{1}{4} \sin \pi \right]$$

$$= \frac{\pi}{2} \left[\frac{\pi}{4} - \frac{1}{4} \sin \pi \right] = \frac{\pi}{2} \left[\frac{\pi}{4} - \frac{1}{4} \sin 0 \right]$$

$$= \frac{\pi}{2} \left[\frac{\pi}{4} - \frac{1}{4} \sin \pi \right]$$

$$= \frac{\pi^2}{8} \text{ units}^3$$

(c)

$v = 5x + 3$, where v is velocity, x is displacement, t is time.

$$\frac{dx}{dt} = 5x + 3$$

$$\int dt = \int \frac{dx}{5x+3}$$

$$t = \frac{1}{5} \ln(5x+3) + C,$$

At $t=0$, $x=2$,

$$0 = \frac{1}{5} \ln(5 \times 2 + 3) + C$$

$$\therefore C = -\frac{1}{5} \ln 13$$

$$\therefore x = \frac{1}{5} \ln(5x+3) - \frac{1}{5} \ln 13$$

$$= \frac{1}{5} \ln \frac{5x+3}{13}$$

$$5x+3 = \ln \frac{5x+3}{13}$$

$$\frac{5x+3}{13} = e^{\frac{5x}{13}}$$

$$5x+3 = 13e^{\frac{5x}{13}}$$

$$5x = 13e^{\frac{5x}{13}} - 3$$

$$\therefore x = \frac{1}{5} (13e^{\frac{5x}{13}} - 3)$$

(d) Number of ways

for the 4 people, 4 people on this side and 4 others

$$= 8 \times 7 \times 6 \times 5$$

Number of ways

for the couple

$$= 7 \times 2$$

couple and 6 others

Number of ways of arranging the other 10

$$= 10!$$

Number of ways of seating the 16 people

$$= (8 \times 7 \times 6 \times 5) \times (7 \times 2) \times 10!$$

$$= 23520 \times 10!$$

(e) (i) $x = 2 \cos(3t + \frac{\pi}{3})$

At $t=0$, $x = 2 \cos \frac{\pi}{3} = 1$

When $x=0$, $2 \cos(3t + \frac{\pi}{3}) = 0$

$$\cos(3t + \frac{\pi}{3}) = 0$$

$$3t + \frac{\pi}{3} = \frac{\pi}{2}$$

$$3t = \frac{\pi}{6}$$

$$t = \frac{\pi}{18}$$

Particle starts at $x=1$ and moves towards the origin.

Distance travelled by the particle when it first reaches the origin

$$= 1 \text{ m}$$

(ii) $x = 2 \cos(3t + \frac{\pi}{3})$

$$\frac{dx}{dt} = \left[-2 \sin(3t + \frac{\pi}{3}) \right] \times 3$$

$$= -6 \sin(3t + \frac{\pi}{3})$$

maximum speed

= maximum of $\left| \frac{dx}{dt} \right|$

$$= \text{maximum of } \left| -6 \sin(3t + \frac{\pi}{3}) \right|$$

$$= 6 \times \text{maximum of } \left| \sin(3t + \frac{\pi}{3}) \right|$$

$$= 6 \times 1$$

$$= 6 \text{ m/s}$$

P.11

$$(f) T = 23 - Ae^{-0.04t}$$

$$\text{At } t=0, T = -18,$$

$$-18 = 23 - Ae^{-0.04 \times 0}$$

$$-18 = 23 - Ae^0$$

$$-18 = 23 - A$$

$$A = 23 + 18$$

$$= 41$$

$$\therefore T = 23 - 41e^{-0.04t}$$

$$\text{When } T = -12,$$

$$-12 = 23 - 41e^{-0.04t}$$

$$41e^{-0.04t} = 23 + 12$$

$$= 35$$

$$e^{-0.04t} = \frac{35}{41}$$

$$0.04t = \ln\left(\frac{41}{35}\right)$$

$$t = \frac{1}{0.04} \ln\left(\frac{41}{35}\right)$$

$$= 3.9556 \dots \text{ minutes}$$

$$= 3.956 \text{ minutes (3 dp)}$$

Question 13

P.12

(a) Prove by Mathematical Induction that $n^3 + 2n$

is divisible by 3 for all integers $n \geq 1$.

Proof:

$$\text{For } n=1, n^3 + 2n = 1^3 + 2 \times 1$$

$$= 3$$

$$= 3 \times 1$$

which is divisible by 3.

The statement is true for $n=1$.

For $n=k$, assume $k^3 + 2k$ is divisible by 3,

i.e., $k^3 + 2k = 3p$, where p is an integer.

For $n=k+1$,

$$(k+1)^3 + 2(k+1)$$

$$= k^3 + 3k^2 + 3k + 1 + 2k + 2$$

$$= (k^3 + 2k) + 3k^2 + 3k + 3$$

$$= (k^3 + 2k) + 3(k^2 + k + 1)$$

$$= 3p + 3(k^2 + k + 1)$$

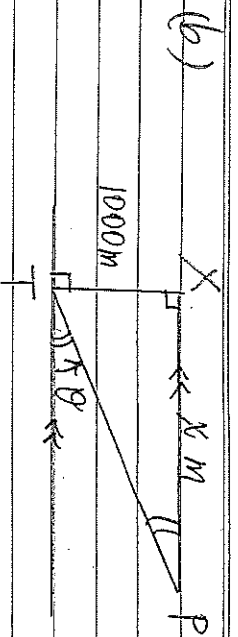
(By assumption)

$$= 3\{p + (k^2 + k + 1)\}$$

which is divisible by 3 since p, k are integers, $\therefore p + (k^2 + k + 1)$ is an integer.

The statement is true for $n=k+1$.

By the Principle of Mathematical Induction, $n^3 + 2n$ is divisible by 3 for all integers $n \geq 1$.



(b) In $\triangle PXT$, $\tan \theta = \frac{1000}{x}$

Differentiate $\tan \theta = \frac{1000}{x}$ w.r.t. t ,

$$\sec^2 \theta \frac{d\theta}{dt} = -\frac{1000}{x^2} \frac{dx}{dt} \quad \text{--- (1)}$$

$$PT^2 = PX^2 + TX^2 \quad (\text{By Pythagoras' thm})$$

$$PT^2 = x^2 + 1000^2$$

$$\sec^2 \theta = \frac{PT^2}{PX^2} = \frac{x^2 + 1000^2}{x^2}$$

$$\frac{dx}{dt} = 200 \text{ ms}^{-1}$$

Put $\sec^2 \theta$, $\frac{dx}{dt}$ into (1),

$$\frac{x^2 + 1000^2}{x^2} \frac{d\theta}{dt} = -\frac{1000}{x^2} \times (200)$$

$$\therefore \frac{d\theta}{dt} = \frac{1000 \times 200}{x^2} \times \frac{x^2}{x^2 + 1000^2}$$

$$= \frac{1000 \times 200}{x^2 + 1000^2}$$

$$= \frac{10^3 \times 2 \times 10^2}{x^2 + (10^3)^2}$$

$$= \frac{2 \times 10^5}{x^2 + 10^6} \quad \text{--- (2)}$$

(ii) When $\theta = \frac{\pi}{4}$, $PX = TX = 1000 \text{ m} = x$

Put $x = 10^3$ into (2) in (1),

$$\frac{d\theta}{dt} = \frac{2 \times 10^5}{(10^3)^2 + 10^6}$$

$$= \frac{2 \times 10^5}{2 \times 10^6}$$

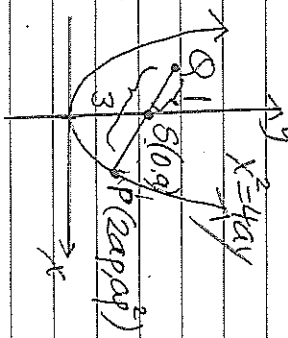
$$= \frac{1}{10} / \text{s}$$

$$= \frac{1}{10} \times \frac{180^\circ}{\pi} / \text{s}$$

$$= 5.729 \dots / \text{s}$$

$$= 6^\circ / \text{s} \quad (\text{nearest degree/s})$$

(c) (i) Q divides PS externally in the ratio 3:1.



$$Q = \left(\frac{3 \times 0 + (-1)2ap}{3 + (-1)}, \frac{3a + (-1)ap^2}{3 + (-1)} \right), \quad 3 \rightarrow S$$

$$= \left(\frac{-2ap}{2}, \frac{3a - ap^2}{2} \right)$$

$$= (-ap, \frac{3a - ap^2}{2})$$

(ii) Let Q be (x_q, y_q)

$$x_q = -ap \quad \text{--- (1)}$$

$$y_q = \frac{3a - ap^2}{2} \quad \text{--- (2)}$$

From (1), $p = -\frac{x_q}{a}$

Put p into (2),

$$y_q = \frac{3a - a(-\frac{x_q}{a})^2}{2}$$

$$y_0 = \frac{3a - \frac{x_0^2}{a}}{2}$$

$$= \frac{3a^2 - x_0^2}{2a}$$

$$2ay_0 = 3a^2 - x_0^2$$

$$x_0^2 = 3a^2 - 2ay_0$$

$$= -2a(y_0 - \frac{3a}{2})$$

As P varies, the locus of Q is a parabola with equation

$$x^2 = -2a(y - \frac{3a}{2})$$

(iii) The locus of Q is

$$x^2 = -2a(y - \frac{3a}{2})$$

$$= -4(\frac{a}{2})(y - \frac{3a}{2})$$

vertex $= (0, \frac{3a}{2})$, Put $y=0$,

$$x^2 = -2a(-\frac{3a}{2})$$

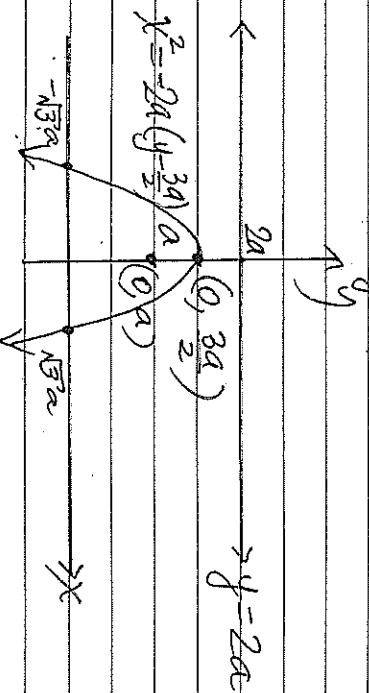
focal length $= \frac{a}{2}$

$$= 3a^2$$

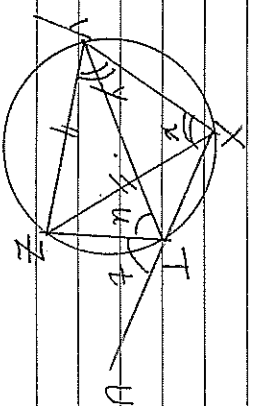
Focus $= (0, a)$

$$x = \pm \sqrt{3}a$$

directrix: $y = 2a$



(d)



Let $\angle XYZ = y$, $\angle XZE = x$, $\angle YTZ = u$, $\angle UTZ = t$.

Prove: TZ bisects $\angle UTY$.

In $\triangle XYZ$, $XZ = YZ$ (given)

$\therefore x = y$ (base angles of isosceles $\triangle XYZ$)

$x = u$ (Ls in the same segment)

$y = t$ (interior $\angle =$ exterior)

$\therefore$ cyclic quadrilateral $(XYZT)$

$\therefore x = y = u = t$

hence $\angle t = t$ i.e., TZ bisects $\angle UTY$.

(a)(i) $2 \cos 3\theta = 1$

$\cos 3\theta = \frac{1}{2}$

$3\theta = 2n\pi \pm \frac{\pi}{3}, n \text{ is integer.}$

$\theta = \frac{2n\pi}{3} \pm \frac{\pi}{9}, n \text{ is integer.}$

(ii) $2 \cos 3\theta = 1, 0 \leq \theta \leq \pi$

$\theta = \frac{2n\pi}{3} \pm \frac{\pi}{9}, n=0, 1, \dots$

$\theta = \pm \frac{\pi}{9}, \frac{2\pi}{3} \pm \frac{\pi}{9}, \dots$

For $0 \leq \theta \leq \pi$,

$\theta = \frac{\pi}{9}, \frac{2\pi}{3} + \frac{\pi}{9}, \frac{2\pi}{3} - \frac{\pi}{9}$

$= \frac{\pi}{9}, \frac{7\pi}{9}, \frac{5\pi}{9}$

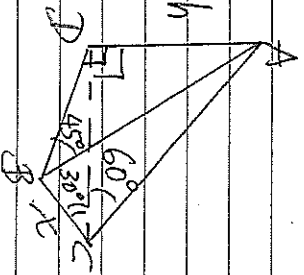
$\therefore \theta = \frac{\pi}{9}, \frac{5\pi}{9}, \text{ or } \frac{7\pi}{9}$

(b)(i) In $\triangle ABD$, $\tan 45^\circ = \frac{BD}{AD}$

$BD = h$

In $\triangle ACD$, $\tan 60^\circ = \frac{h}{CD}$

$\therefore CD = \frac{h}{\sqrt{3}}$



In $\triangle BCD$

$BD^2 = CD^2 + BC^2 - 2 \times CD \times BC \cos \angle BCD$

$h^2 = \frac{h^2}{3} + x^2 - 2 \times \frac{h}{\sqrt{3}} \times x \cos 30^\circ$

$= \frac{h^2}{3} + x^2 - \frac{2hx}{\sqrt{3}} \times \frac{\sqrt{3}}{2}$

$h^2 = \frac{h^2}{3} + x^2 - hx$

$3h^2 = h^2 + 3x^2 - 3hx$

$2h^2 + 3xh - 3x^2 = 0$

(ii) $2h^2 + 3xh - 3x^2 = 0$

Divide by x^2 , $\frac{2h^2}{x^2} + \frac{3xh}{x^2} - 3 = 0$

$2\left(\frac{h}{x}\right)^2 + 3\left(\frac{h}{x}\right) - 3 = 0$

$\frac{h}{x} = \frac{-3 \pm \sqrt{3^2 - 4 \times 2 \times (-3)}}{2 \times 2}$

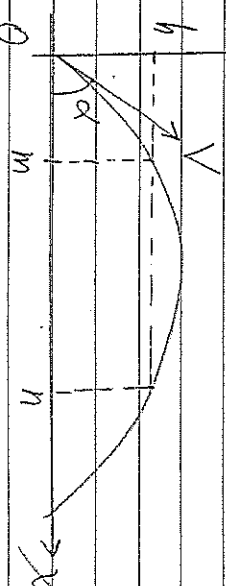
$= \frac{-3 \pm \sqrt{33}}{4}$

$\therefore \frac{h}{x} = \frac{-3 + \sqrt{33}}{4}$

since $h > 0, x > 0$,

$\frac{h}{x} = \frac{\sqrt{33} - 3}{4}$ reject $\frac{-3 - \sqrt{33}}{4}$

(c)(i)



For horizontal motion $\ddot{x} = 0$

$$\dot{x} = C_1$$

$$\text{At } t=0, \quad \dot{x} = V \cos \alpha$$

$$\therefore C_1 = V \cos \alpha$$

$$\dot{x} = V \cos \alpha$$

$$x = Vt \cos \alpha + C_2$$

$$\text{At } t=0, \quad x=0,$$

$$0 = 0 + C_2$$

$$\therefore C_2 = 0$$

$$x = Vt \cos \alpha$$

$$\text{For Vertical motion, } \ddot{y} = -10$$

$$\dot{y} = -10t + C_3$$

$$\text{At } t=0, \quad \dot{y} = V \sin \alpha$$

$$V \sin \alpha = 0 + C_3$$

$$C_3 = V \sin \alpha$$

$$\therefore \dot{y} = -10t + V \sin \alpha$$

$$y = -5t^2 + Vt \sin \alpha + C_4$$

$$\text{At } t=0, \quad y=0,$$

$$0 = 0 + 0 + C_4$$

$$C_4 = 0$$

$$y = -5t^2 + Vt \sin \alpha$$

$$y = Vt \sin \alpha - 5t^2$$

$$(11) \quad x = Vt \cos \alpha$$

$$y = Vt \sin \alpha - 5t^2 \quad \text{--- (2)}$$

$$\text{From (1), } t = \frac{x}{V \cos \alpha}$$

$$\text{Put } t = \frac{x}{V \cos \alpha} \text{ into (2),}$$

$$y = V \sin \alpha \times \frac{x}{V \cos \alpha} - 5 \left(\frac{x}{V \cos \alpha} \right)^2$$

$$= x \tan \alpha - \frac{5x^2}{V^2 \cos^2 \alpha}$$

$$\text{Put } x=m, \quad y=h,$$

$$\therefore h = m \tan \alpha - \frac{5m^2}{V^2 \cos^2 \alpha}$$

$$\frac{5m^2}{V^2 \cos^2 \alpha} = m \tan \alpha - h$$

$$V^2 \cos^2 \alpha = \frac{5m^2}{m \tan \alpha - h}$$

$$V^2 = \frac{5m^2 \times \frac{1}{\cos^2 \alpha}}{m \tan \alpha - h}$$

$$= \frac{5m^2 \sec^2 \alpha}{m \tan \alpha - h} \quad \text{--- (3)}$$

$$= \frac{5m^2 (1 + \tan^2 \alpha)}{m \tan \alpha - h} \quad \text{--- (By } \sec^2 \alpha = 1 + \tan^2 \alpha)$$

$$(111) \text{ Similarly put } x=n, y=h \text{ into}$$

$$y = x \tan \alpha - \frac{5x^2}{V^2 \cos^2 \alpha}$$

$$h = n \tan \alpha - \frac{5n^2}{V^2 \cos^2 \alpha}$$

$$V^2 = \frac{5n^2 (1 + \tan^2 \alpha)}{n \tan \alpha - h} \quad \text{--- (4)}$$

Equating (3) and (4),

$$5m^2(1+\tan^2\alpha) = 5n^2(1+\tan^2\alpha)$$

$$m\tan\alpha - h = \frac{m^2}{n^2} n\tan\alpha - h$$

$$\frac{m\tan\alpha - h}{n^2} = \frac{n\tan\alpha - h}{n^2}$$

$$m^2(n\tan\alpha - h) = n^2(m\tan\alpha - h)$$

$$m^2n\tan\alpha - m^2h = mn^2\tan\alpha - n^2h$$

$$(m^2n - mn^2)\tan\alpha = m^2h - n^2h$$

$$mn(m-n)\tan\alpha = (m^2 - n^2)h$$

$$\tan\alpha = \frac{(m^2 - n^2)h}{mn(m-n)}$$

$$= \frac{(m-n)(m+n)h}{mn(m-n)}$$

$$= \frac{h(m+n)}{mn}$$